

Subject2Image

Evaluating Adaptation Approaches for Subject-Driven Image Generation

Sofia Caruso, Giacomo Tessari
Computer Vision and Cognitive Systems

Introduction

Can AI Learn Your Style Without Forgetting Everything Else?

In this work, we systematically evaluate subject-driven personalization across three distinct pipelines, **comparing Low-Rank Adaptation (LoRA) and Textual Inversion**.

We benchmark all implementations on the standard DreamBooth dataset using a comprehensive suite of quantitative metrics:

- CLIP-T
- CLIP-I and DINO-I
- LPIPS

Experimental results demonstrate that:

- LoRA-based weight adaptation significantly outperforms Textual Inversion
- The first LoRA pipeline achieves the highest overall performance in subject fidelity and visual preservation

Method and Implementation

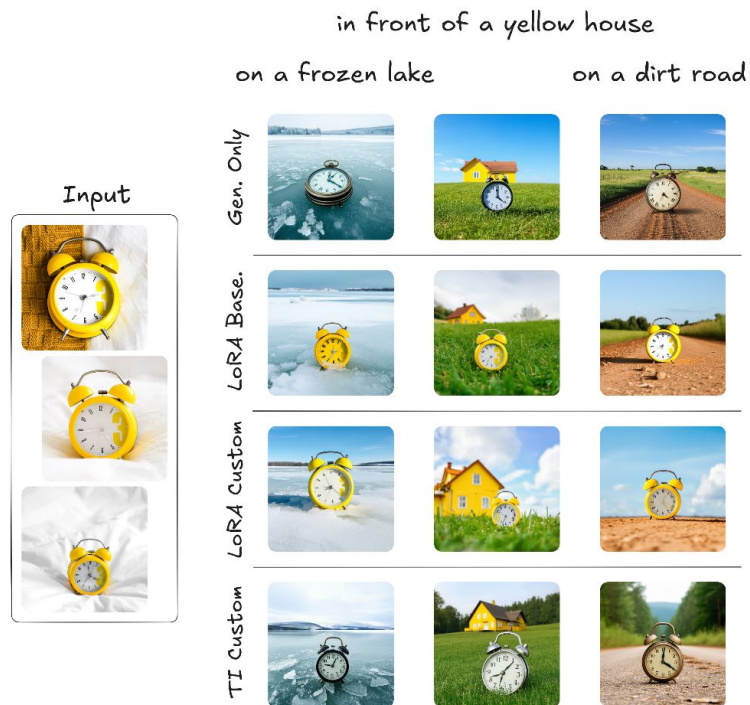
3 pipelines:

- **Generation Only:** An unpersonalized generation baseline, used as an anchor to measure the base model's intrinsic text-prompt alignment without subject adaptation.
- **LoRA Baseline:** A parameter-efficient fine-tuning setup that integrates Low-Rank Adaptation into the DreamBooth framework.
- **LoRA Custom:** An extended variant of the first pipeline, combining LoRA-based weight adaptation with spatial data augmentation on the fine-tuning reference set.
- **Textual Inversion Custom:** A modern adaptation of Textual Inversion scaled to the Stable Diffusion 3.5 architecture, optimizing text embeddings within its multi-encoder prompt space.

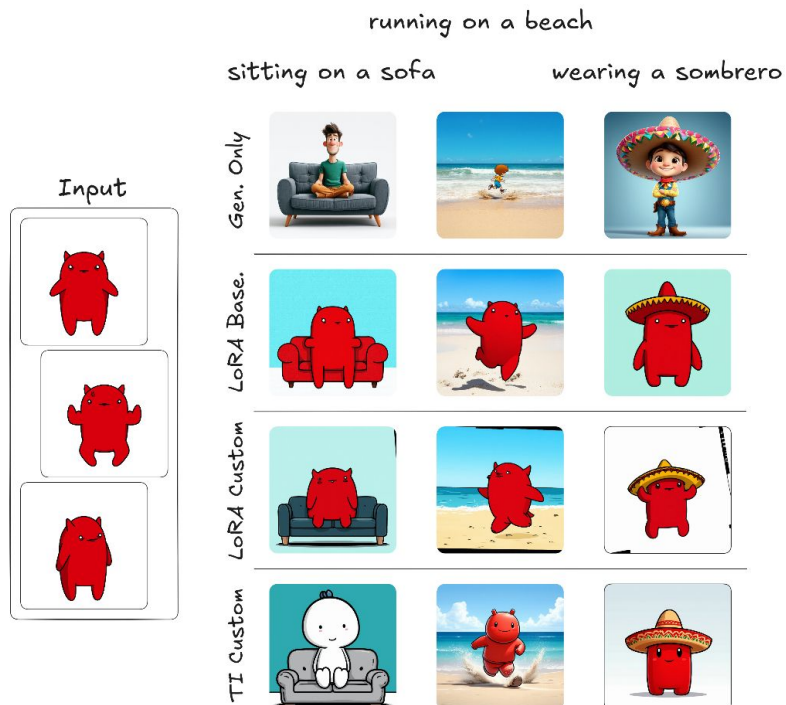
Quantitative results

	CLIP-T $\uparrow$	CLIP-I $\uparrow$	CLIP-I $\times$ CLIP-T $\uparrow$	DINO-I $\uparrow$	LPIPS $\downarrow$
Generation Only	0.8953	0.61	0.5461	0.2846	0.7919
LoRA Baseline	0.8678	0.6509	0.5648	0.6303	0.7134
LoRA Custom	0.8723	0.6451	0.5627	0.6252	0.7201
TI Custom	0.8763	0.6175	0.5411	0.3016	0.7780

Qualitative results



Qualitative results



Discussion

- **Weight-Space vs. Text-Space:** LoRA fine-tuning demonstrates a fundamental capacity advantage over Textual Inversion to capture complex 3D geometry and fine-grained textures. In contrast, embedding-only optimization lacks the capacity to retain instance details, yielding weak identity preservation.
- **Effect of Spatial Data Augmentation:** Applying spatial transformations like cropping and rotation marginally improves prompt adherence, but degrades subject fidelity. These geometric perturbations disrupt high-frequency pixel alignments essential for reproducing fine details. Moreover, reference data augmentation causes the model to be more prone to overfitting.
- **Multi-Encoder Complexity:** The underperformance of Textual Inversion is largely driven by the architectural complexity of Stable Diffusion 3.5, which integrates a triple text-encoder system with an MM-DiT backbone. Optimizing embeddings alone without fine-tuning the backbone's text stream provides insufficient control to steer generation within practical training limits.